

Quiz 12

A politician pays for a poll before an election. Although she doesn't know it, she is the pick of 47% of the voters for the election.

Suppose her poll will have a random sample of 500 voters. You may treat the preferences of people in the sample as independent.

1. Let N be the random variable representing the number of voters in the sample who pick this politician for the election. Explain how to write N as a sum of 500 indicator random variables. You must define your 500 events and the random variables clearly.

Let A_i ($i = 1, \dots, 500$) be the event that voter i in the sample picks the politician. Let X_i be the indicator of A_i . Then $N = X_1 + \dots + X_{500}$.

2. Use your expression for N to calculate $E[N]$ and $\text{Var}(N)$. Justify your calculations briefly.

$$E[N] = E[X_1] + \dots + E[X_{500}] = 0.47 + \dots + 0.47 = (500)(0.47) = 235$$

Justification:

Since expected value is additive, we can sum the expected values of the X_i to get the expected value of N . Each X_i has the Bernoulli(0.47) distribution, and so $E[X_i] = 0.47$.

$$\text{Var}(N) = \text{Var}(X_1) + \dots + \text{Var}(X_{500}) = (0.47)(0.53) + \dots + (0.47)(0.53) = 500(0.47)(0.53) = 124.55$$

Justification:

We treat the preferences of people in the sample as independent. Therefore, $\text{Var}(N)$ is the sum of the variances of the X_i . Each X_i has the Bernoulli(0.47) distribution, and so $\text{Var}(X_i) = (0.47)(0.53)$.

3. Show how to use the Central Limit Theorem to estimate the probability that at least half of the people in the sample pick this candidate for the election.

We estimate $P(N \geq 250)$ using the Central Limit Theorem. Since N is a discrete random variable, we use the histogram correction and estimate $P(N \geq 249.5)$. The z-score for 249.5 is $\frac{249.5-235}{\sqrt{124.55}} \approx 1.30$.

$$\begin{aligned} P(N \geq 249.5) &= P\left(\frac{N - 235}{\sqrt{124.55}} \geq 1.30\right) \\ &= 1 - P\left(\frac{N - 235}{\sqrt{124.55}} < 1.30\right) \\ &\approx 1 - P(\text{standard normal} < 1.30) \\ &= 1 - P(\text{standard normal} \leq 1.30), \text{ since the normal distribution is continuous} \\ &\approx 1 - 0.9032 = 0.0968. \end{aligned}$$

The chance of having a sample of 500 voters with at least 50% supporters for this candidate is a bit less than 10%.